Automatic Parking Gate System

COAL Course Project

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Project Overview



THIS PROJECT PRESENTS AN AUTOMATIC PARKING GATE SYSTEM USING ARDUINO AND A SERVO MOTOR.



THE SYSTEM DETECTS VEHICLES AND AUTOMATICALLY OPENS OR CLOSES THE GATE WITHOUT HUMAN INVOLVEMENT.



IT DEMONSTRATES CORE CONCEPTS OF MICROCONTROLLERS AND COAL.

Problem Statement



Manual parking systems are slow, inefficient, and require constant human effort.



Commercial automated systems are expensive.



This project provides a low-cost educational solution using basic electronic components.

Project Objectives

- Automate parking gate operation
- Detect vehicles using sensors
- Control gate using servo motor
- Apply COAL and microcontroller concepts
- Build a low-cost, functional prototype

System Architecture

- Sense – Ultrasonic sensor detects vehicle
- Think – Arduino processes distance data
- Act – Servo motor opens or closes the gate
- Threshold distance: 30 cm

Operational Workflow

1. System initializes and calibrates sensor

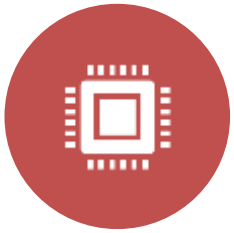
2. Ultrasonic sensor measures distance

3. Arduino compares distance with threshold

4. Gate opens if vehicle detected

5. Gate closes after delay

Software Requirements



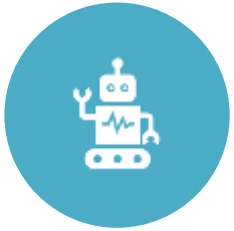
- Arduino IDE (C++)



- Servo.h library



- Optional: Fritzing for circuit diagrams



Software follows a continuous Sense-Think-Act loop.

Hardware Requirements



- Arduino Uno (ATmega328P)



- HC-SR04 Ultrasonic Sensor



- SG90 Servo Motor



- Power supply (5V–9V)



- Jumper wires, breadboard, gate arm

Challenges



- Power management for servo motor



- Ultrasonic sensor calibration



- Accurate timing for gate movement

Future Improvements

- RFID or IR access control
- IoT and mobile app monitoring
- Advanced sensors like LiDAR

Conclusion

- The project successfully demonstrates an automatic parking gate system.
- It integrates sensors, actuators, and control logic effectively.
- The system is suitable for learning embedded systems and COAL concepts.





THANK YOU